

Ananta Narayan Shrestha

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Research Interests

Program analysis and software testing; AI for software engineering, particularly evaluating and validating LLM-generated code changes; security tooling for production systems (fuzzing, static analysis); data-intensive and high-performance computing infrastructure.

Education

Arizona State University Aug 2022 – May 2024
M.S., Computer Science GPA 4.00/4.00 Tempe, AZ

- Coursework: Foundations of Machine Learning, Artificial Intelligence, Knowledge Representation, Cloud Computing, Data Visualization, Software Project Management.

Motilal Nehru National Institute of Technology (MNNIT) Jun 2014 – May 2018
B.Tech., Computer Science & Engineering Prayagraj, India

Publications

1. Powell, B., Shrestha, A., et al. “Where are all the Electric Vans?” ASCE International Conference on Transportation and Development (ICTD), 2025.
2. Venkat, N.D., Dewang, R.K., Sundar, B., Agrawal, A., Shrestha, A.N., Tiwari, A. “An Android Application for Automatic Content Summarization of News Articles Using Multilayer Perceptron.” Proceedings of ICBMA 2019, Lecture Notes in Networks and Systems, vol. 150, Springer. doi:10.1007/978-981-15-8377-3_32

Research Experience

Research Software Engineer Intern Aug 2023 – Sep 2025
National Renewable Energy Laboratory (NREL) Golden, CO

- Built and validated a 1 TB data-processing pipeline for the national EV infrastructure grid simulation on HPC (Python, Polars, Pydantic, Parquet), reducing execution time from 10 hours to 30 minutes. Architected a Pydantic validation layer enforcing strict configuration schemas.
- Made the OpenPATH public dashboard testable. Its chart logic called `get_ipython().run_line_magic()` at import time, so no test could import the module. Decoupled the IPython dependency to be runtime-detected and optional, then built the CI harness that did not previously exist (GitHub Actions → docker-compose → pytest with coverage), replacing a five-step manual verification ritual. Design tradeoffs documented publicly in `em-public-dashboard#171`, including why notebook-level testing was rejected and why automated chart-image comparison was scoped out as unsolved.
- Redesigned the OpenPATH public dashboard (Python, JavaScript, MongoDB) serving 5,000+ users across 34 deployments; replaced pie charts with 100% stacked bar charts for cross-program comparability.
- Contributed to RouteE Compass, an open-source energy-aware routing engine in Rust enabling Google Maps’ eco-routing. Recognized with the U.S. DOE Vehicle Technologies Office AMR Team Award, 2025.
- Built a machine-learning pipeline (random forest with feature engineering and cross-validation) for deployment-stage commute-mode inference over large-scale mobility datasets.
- Teams: e-mission/OpenPATH, RouteE Compass, EVI-Pro, EVI-Grid National Framework, On-Demand Transit Research.

Project Team Leader (graduate research project) Jan 2023 – May 2023
Arizona State University Tempe, AZ

- Led a team of seven graduate students on information assurance and security in machine learning, targeting malicious URL detection in social networks; investigated DCNN and CNN approaches.

Open-Source Contributions

- e-mission/em-public-dashboard: 33 pull requests authored, 22 merged, covering chart, metrics and colour work plus the test-automation effort described above.
- NatLabRockies/routee-compass (Rust): four merged pull requests, the largest (#286) moving JSON getter extensions into the core crate across 40 files; also SI energy-rate units (#280), a TOML inject format (#285), and gzip-compressed query files (#483). Plus conda-forge feedstock #14 for `osx-arm64`.
- nanvix/nanvix (Rust): six commits merged to the default branch of Microsoft Research’s multikernel OS for serverless sand-boxing: error-type definitions in the kernel `sys` layer and build-script hardening.

Industry Experience

Software Engineer

Dec 2025 – Present

Exito Tech Inc.

New York, NY

- Building an AI chatbot on AWS using Anthropic Claude and retrieval-augmented generation to improve response grounding and quality.

Software Engineer Intern II

May 2023 – Aug 2023

Yahoo Inc., Edge Infrastructure

Champaign, IL

- Integrated four fuzzing engines (Google FuzzTest, libFuzzer, AFL++, Honggfuzz) into Yahoo's large-scale production DNS infrastructure in C++17, identifying and reporting two critical memory-management vulnerabilities.
- Automated the fuzzing workflow with CI/CD pipelines (ScrewDriver, Makefile, CMake) in a Dockerized Linux environment; authored a ~50-page engineering report on engine tradeoffs and LLVM/OpenStack integration.

Software Engineer I & II

Jun 2018 – Jun 2022

Philips India Ltd., Ultrasound R&D

Bangalore, India

- Architected a multi-threaded, low-latency (5 ms) real-time audio processing module (Swift, C++, AVFoundation/AVAudioEngine) for Pulse Wave Doppler playback on the Lumify iOS app, under an FDA-regulated release process.
- Rhythm Service Platform (C++, MVVM): designed multi-level user login using a publish-subscribe pattern and layered architecture, maintaining >95% platform-layer and >85% application-layer unit test coverage.
- Triaged and resolved critical multi-threading and connectivity defects in a 20 GB legacy C++ codebase; built a C++/C# static-analysis CLI tool that identified stale parameters, resolving ten high-priority production incidents.
- Designed a secure multi-level authentication portal (C++, C#, WPF/MVVM) incorporating AES and SHA-256. Shipped ultrasound features (Quad AFI, High Q Doppler, 3D Preset within Preset) reducing exam execution time by up to 40%.

Selected Projects

- Energy Policy RAG (github.com/iantei/energy-rags) : retrieval-augmented generation system with a Medallion (bronze/silver/gold) ETL on Polars and Parquet, Docling PDF parsing, ChromaDB, LangChain, and Dockerized GPU inference via Ollama; 18-test integration suite.
- Realtime Audio Lab (github.com/iantei/swiftui-realtime-audio) : lock-free single-producer/single-consumer ring buffer with acquire/release atomics and wait-free parameter passing, so the CoreAudio render thread never allocates or locks under a ~5.3 ms deadline (48 kHz / 256-frame buffer). Windowed real FFT via Accelerate/vDSP; Swift 6 strict concurrency. Tests cover ring-buffer wrap-around, torn reads, and oscillator phase continuity. Runs as a macOS host application.
- Distributed Banking System: client-centric consistency models (Read-Your-Writes, Monotonic Writes) and Lamport logical clocks for causal event ordering across replicated servers (Python, gRPC).

Technical Skills

Languages: Python, C++, Rust, Swift, Java, C#, JavaScript, SQL, C

Analysis & Security: fuzzing (Google FuzzTest, libFuzzer, AFL++, Honggfuzz), static analysis, vulnerability identification, AES/SHA-256

Data & HPC: Polars, Parquet, PostgreSQL, MongoDB, ChromaDB, Slurm, distributed transactions and consistency

ML & GenAI: scikit-learn, PyTorch, Anthropic API, LangChain, RAG architecture, Ollama

Tooling: Docker, Docker Compose, GitHub Actions (CI/CD), pytest, CMake, AWS (EC2, S3, SQS, Lambda, IAM, ECR)

Awards and Honors

- U.S. Department of Energy, Vehicle Technologies Office: AMR Team Award, 2025 (RouteE & FASTSim, enabling Google Maps eco-routing).
- Philips Ultrasound Hackathon: Winner 2019–20 (High Q Doppler Trace Enabler, 200 participants); Runner-up 2018–19 (Unified Defect Writing Tool, 150 participants).
- Philips Planning Excellence Award: Winner, Q4 2021, cited for quality, predictability and productivity; Team Winner, Q1 2021; Q3 2022.

References

Available on request.